(Please write your Enrollment Number)

Enrollment No. 06001197023

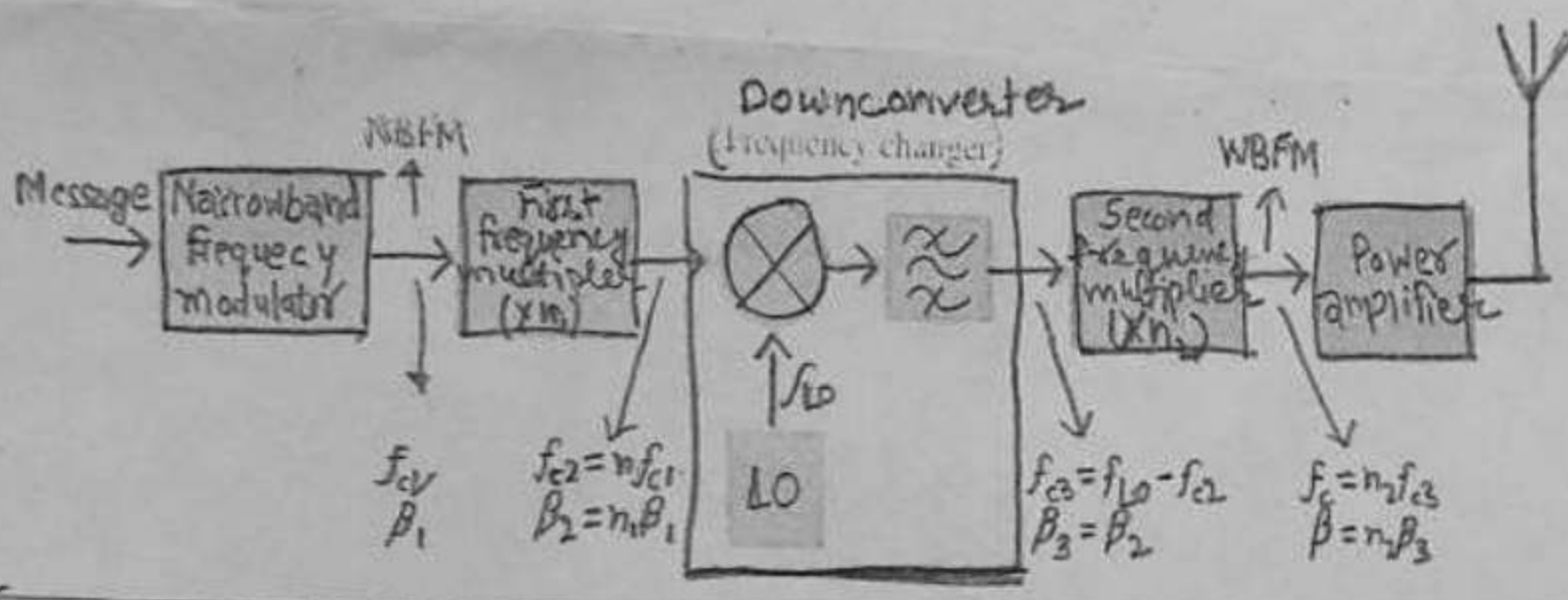
End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<3>
(December, 2024)

Subject Code: BEC 211	
Time :3 Hours	Subject: Communication System
	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1			(2.5*8=20)
	a)	A stationary random process X(t) is applied to the input of a system for which $h(t) = t^2e^{-4t}u(t)$. If E[X(t)] = 5. What is the mean value of the system's response Y(t)?	
	b)	If X(t) and Y(t) are two random process with autocorrelation function $R_{XX}(t)$ and $R_{YY}(t)$ respectively then prove that $\lvert R_{XY}(\tau)\rvert \le \sqrt{(R_{XX}(0)R_{YY}(0))}$	
	c)	Define bandlimited white noise.	
	d)	With the aid of the block diagram explain the principle and working of Super hetrodyne receiver.	
	e)	Explain the working principle of balanced slope discriminator used for FM detection.	
	f)	How AM is generated using square law modulator? Derive relevant expressions.	
	g)	What is capture effect and threshold effect in Angle modulation?	
	h)	A carrier wave of frequency 10 MHz and peak value 10V is amplitude modulated by a 5 KHz sine wave of amplitude 6V. Determine the modulation index and amplitude of the side frequencies.	
		UNIT I	
Q2	a)	Consider a random process $Y(t)= X(t)\cos(\omega_0 t + \theta)$ where X(t) is a wide-sense stationary random process, θ is a random variable independent of X(t) and is distributed uniformly in $(-\pi, \pi)$ and w_0 is a constant. Prove that Y(t) is wide-sense stationary.	(5+5)
	b)	X is a random variable with mean $\mu_X = 3$, $\sigma_X^2 = 2$. (i) Determine the second-order moment of X about the origin. (ii) Determine the mean of random variable Y where Y = –6X + 22	
Q3	a)	The probability density functions of statistically independent random variables X and Y are $f_X(x) = \frac{1}{2}e^{-(x-1)/2}u(x-1)$; $f_Y(y) = \frac{1}{4}e^{-(y-3)/4}u(y-3)$. Find the pdf of Z=X+Y	(5+5)
	b)	The joint pdf of random variables (X, Y) is $f_{X,Y}(x,y) = \frac{1}{4}e^{-\lvert x\rvert-\lvert y\rvert}$ $-\infty < x \le \infty$, $-\infty < y \le \infty$ (i) Are X and Y statistically independent random variables? (ii) Calculate the probability for X≤ 0 and Y≤ 0.	
		UNIT II	
Q4	a)	Describe the generation of SSB using phase discrimination method	(5+5)

	b) What is envelop detector? Derive the expression for avoiding the diagonal clipping in the envelop detector.	
Q5	a) With necessary circuit diagram and waveforms, explain how DSB-SC wave is generated using Balance Modulators b) A square-law demodulator, where the input voltage $v_{am}(t)$ is the incoming AM signal. For small input voltage levels, the diode current is a nonlinear function of $v_{am}(t)$ so that the output voltage $v_o(t)$ is given approximately by $v_o = \alpha_1 v_{am} + \alpha_2 v_{am}^2$ (i) Evaluate the output voltage $v_o(t)$ and show that it contains the original message signal. (ii) Discuss how the message signal may be extracted from $v_o(t)$ and the conditions that must be satisfied to minimise distortion	(5+5)
	UNIT III	
Q6	a) Derive the NBFM and WBFM expression. Explain the Armstrong method of generation of WBFM.	(5+5)
	b) Figure shows the block diagram of an Armstrong wideband FM transmitter that employs two stages of frequency multiplication. The message is a sinusoid of frequency f_m = 15 kHz and the NBFM signal has carrier frequency f_{c1} = 100 kHz and frequency deviation f_{d1} = 750 Hz. The required transmission from the power amplifier stage is a WBFM signal having carrier frequency f_c = 120 MHz and frequency deviation f_d = 75 kHz. The first frequency multiplication factor is n_1 = 5.Determine: (i) The required second frequency multiplication factor n_2. (ii) The required LO frequency f_{LO}. (iii) The (Carson's) bandwidth of the FM signal at the output of each processing block in the transmitter	



Q7	a) Explain the direct method of generation of FM b) Explain Ratio detector. Compare it with foster seelay discriminator.	(5+5)
	UNIT IV	
Q8	a) An amplifier has a standard spot noise figure F_0 = 10 dB. An engineer uses the amplifier to amplify the output of antenna whose temperature T_s = 200 K (i) What is the effective input noise temperature of the amplifier? (ii) What is the operating noise figure? b) Derive the figure of merit of DSB-SC modulation scheme.	(5+5)
Q9	a) Two resistors of 50 kW and 100 kW are at room temperature. For a bandwidth of 500 kHz, calculate the thermal noise voltage generated by (i) each resistor, (ii) the two resistors in series, and (c) the two resistors in parallel. b) What is pre-emphasis and de-emphasis.	(5+5)